

Subham Karmakar

Applied AI & Full-Stack Developer | subham12r.me | [GitHub](#) | [LinkedIn](#) | [+91 9434474878](tel:+919434474878) | rikk4335@gmail.com

SUMMARY

Applied AI and full-stack developer and Computer Science student building production-oriented AI, real-time, web, desktop, backend, and ML systems. Experienced with LLM and RAG systems, Next.js, React, TypeScript, Python/FastAPI, Go, PostgreSQL, Redis, Docker, vector search, persistent memory, and end-to-end product delivery.

EDUCATION

Adamas University - Bachelors in Technology, Computer Science and Engineering

CGPA: 9.0/10

Aug 2024 - Jul 2028 (Expected) , Kolkata, India

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, C++, SQL

Frontend & Backend: React, Next.js, Tailwind CSS, Electron, React Native (Expo Go), Node.js, Express, FastAPI, REST APIs, WebSockets, SSE

Data & Infrastructure: PostgreSQL, SQLite, Redis, pgvector, sqlite-vec, Git, Docker, Docker Compose, GitHub Actions, Nginx, Linux

AI & ML: OpenAI, Anthropic and Gemini APIs, OpenRouter, Ollama, LangChain, RAG, semantic chunking, embeddings, vector search, XGBoost, Random Forest, scikit-learn

EXPERIENCE

Ryze AI - Frontend Developer Intern

Jan 2026 - Apr 2026, Remote

- Shipped production UI components using React, TypeScript, and Tailwind CSS; contributed to code reviews and sprint delivery.
- Integrated REST APIs and implemented component-level caching to improve data-fetching behavior.

Adamas University - Moodle LMS Cloud Maintainer(Part-time)

Nov 2025 - May 2026, Kolkata, India

- Administered a cloud-hosted Moodle LMS supporting **1,000+ students and faculty**, including system health monitoring and incident response.

SELECTED PROJECTS

CYCODERS - Community Platform | Next.js, TypeScript, Express.js, PostgreSQL, Docker, Nginx | [Live](#)

- Built a community platform serving **600+ active users** with member dashboards, QR-code event check-in, points-based progression, and leaderboards.
- Implemented JWT authentication, role-based access, event and attendance workflows, PostgreSQL persistence, and an Nginx reverse proxy for VPS deployment.

Ruse - Temporary Screen-Sharing Platform | Go, Next.js, TypeScript, Redis, LiveKit, WebSockets, Docker | [Live](#)

- Built temporary rooms supporting screen sharing, camera, microphone, and **ephemeral chat through a Go backend**, Next.js interface, and LiveKit.
- Added owner-approved admission, Redis-backed room state and capacity, hashed session tokens, and **15-minute media access tokens** with publishing permissions.

SecureMailScope - Mail-Transport Security Assessment | Python, FastAPI, XGBoost, Random Forest, scikit-learn, PostgreSQL, Next.js | [Live](#) | [GitHub](#)

- Contributed dashboard, ML, and backend components for an SIH team project assessing **SMTP, IMAP, and POP3** sessions using calibrated classifiers and evidence-linked security rules.
- XGBoost achieved **98.00%** accuracy and **0.9763 macro-F1** on **7,665** held-out synthetic packet-derived sessions; FastAPI combines ML predictions with deterministic findings.

Atlas - AI Desktop Assistant | Electron, TypeScript, Python, FastAPI, SQLite, sqlite-vec, FastEmbed | [GitHub](#)

- Built a desktop assistant with OpenAI, Anthropic, Gemini, OpenRouter, and Ollama adapters, supporting streamed responses, image inputs, and voice transcription.
- Implemented persistent conversational memory with 384-dimensional embeddings, semantic chunking, sqlite-vec retrieval, rolling summaries, and entity-relationship memory.

Noted - Collaborative Notes & Study Platform | Next.js, TypeScript, Node.js, Yjs, Socket.io, PostgreSQL, Redis, Stripe | [Live](#)

- Built collaborative notes with Yjs CRDTs and Socket.io, participant presence, Redis-backed state snapshots, and validated AI actions for notes, folders, and tasks.
- Integrated Stripe Checkout, billing-portal access, and subscription webhooks for paid-plan lifecycle management.

LEADERSHIP, CERTIFICATIONS & ACHIEVEMENTS

- Student Lead, CYCODERS - Lead a 2,000+ member university technology community spanning multiple chapters; previously served in Technical Lead roles across CYCODERS and ACM.
- Smart India Hackathon 2025 - Internal finalist; ranked among the top 10 software teams out of 150 teams.
- Oracle Agentic AI Certified Foundations - Aug 2026
- AWS Academy Graduate - Cloud Foundations and Cloud Architecting - Sep 2026
- Anthropic Education - Claude Code 101 and MCP Introduction - Jun 2026